ORIGINAL RESEARCH ARTICLE

# Yttria-Stabilized Zirconia Assisted Green Electrochemical Preparation of Silicon from Solid Silica in Calcium Chloride Melt

YUNMING GAO, ZHENBIAO HUANG, LIN HE, GEORGE Z. CHEN, QINGWEI QIN, and GUANGQIANG LI

A novel integrated cell with $O^{2-}|YSZ|Pt|O_2(air)$ reference and counter electrodes was constructed using a short yttria-stabilized zirconia solid electrolyte (YSZ) tube. Combining with cyclic voltammetry and potentiostatic electrolysis methods, green electrochemical preparation of Si from solid $SiO_2$ in $CaCl_2$ melt at 1173 K was studied via an experimental apparatus containing the novel integrated cell under completely carbon-free conditions; the effect of electrolysis time on the morphology of the Si product was also investigated by scanning electron microscopy with energy dispersive x-ray spectroscopy (SEM-EDS). The results show that the morphology of the product obtained from potentiostatic electrolysis at a low overpotential (− 1.6 V) undergoes an evolution from $SiO_2$ raw powder with different sizes to aggregates of spherical particles and small particles with partial reduction, and then to Si nuclei, and finally to Si wires or flakes. The morphology of electrolytic products has little relation with that of $SiO_2$ raw powder and may be controlled by applying different potentials. Furthermore, the longer the electrolysis time, the more the Si wires grow, and the higher the Si purity overall. It is feasible that the experimental apparatus without the sealed stainless steel reactor and any carbonaceous materials can be used to prepare Si from solid $SiO_2$ in $CaCl_2$ melt and release $O_2$ gas at the same time.

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## I. INTRODUCTION

SILICON is the most basic functional material in the information, new energy and new materials industries. Two decades ago the introduction of the FFC-Cambridge process generated a continued interest in electrochemical preparation of Si from solid $SiO_2$ in the $CaCl_2$ melt.[1–5] The studies focus not only on improving the purity of the Si product, but also on its preparation with controllable morphology, structure, *etc*.[5–13] It is well known that Si is active at high temperatures. Therefore, a sealable heat resistant stainless steel reactor had to be used to prevent the oxidation of Si in previous works,[2,4,5,9,11,13] and various carbonaceous materials were widely employed as the electrodes or crucible container.[2–15] A large voltage was applied to electrolyze solid $SiO_2$.[5,9,14,15] In addition, an ion isolation membrane should be set up in the electrochemical cell to avoid physical interference of the components in different electrode zones; however, it was difficult to choose a suitable material for the ion membrane at high temperatures. These practices not only lead to low efficiency of electrolysis due to electronic conduction in molten salt, and the release of unfriendly gases such as $CO_x$ or even $Cl_2$, but also induce a side reaction on calcium reduction and the contamination of products by various components including stainless steel and potential impurities such as carbon and calcium.[5,16,17]

$ZrO_2$-doped with oxides such as $Y_2O_3$ or MgO is a kind of oxygen-ion-conducting solid electrolyte with good stability and strong resistance to melting at high temperatures.[18] In recent years, it has been used as a solid oxide membrane (SOM) in the electrochemical

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preparation of various metals or intermetallic compounds (*e.g.,* Mg, Al, Ti, Ca, Ta, Fe, $CeNi_5$, Ti-Fe, $TiAl_3$, *etc.*) from respective metal oxides.[14,15,19–23] Furthermore, it is frequently used to construct an independent $O^{2-}|ZrO_2|Pt|O_2$(air) reference electrode (RE) with flowing air at high temperatures for oxygen-related electrochemical studies.[18,24,25] $O^{2-}|ZrO_2|Pt|O_2$(air) RE is essentially an $O^{2-}/O_2$ pair, which is generally considered to be responsive, stable, reversible and reproducible.[18,25] In view of the above applications, the authors' research group constructed a novel integrated cell with $O^{2-}|ZrO_2|Pt|O_2$(air) RE by using a $ZrO_2$ solid electrolyte tube, which is simple and unique in structure and fabrication. The integrated cell has been successfully applied to electrochemical fundamental research in high-temperature melts.[26–28]

It is well known that solid $SiO_2$ is electronically insulating at high temperatures, and the electroreduction of $SiO_2$ to Si (Eq. [1]) can occur only at three-phase interlines (3 PIs) of the conductive current collector and solid $SiO_2$ cathode and molten salt (electrolyte):[17,29]

$$SiO_2(s) + 4e \rightarrow Si + 2O^{2-} \quad [1]$$

The generated Si nuclei may act as new current collectors due to their good conductivity. With the reduction, the Si nuclei continue to grow and evolve, and the 3 PIs extend until the $SiO_2$ is completely reduced. In this work, Si was attempted to prepare from solid $SiO_2$ in $CaCl_2$ melt at 1173 K in an experimental apparatus containing the novel integrated cell with $O^{2-}|YSZ|Pt|O_2$(air) RE constructed by yttria-stabilized zirconia solid electrolyte (YSZ) tube, and $O_2$ gas was released at the same time. The effect of electrolysis time on the morphology of the electrolytic Si was also investigated at a low overpotential. Herein the YSZ tube was used as the cell container, and also as the ion conducting but electron insulating membrane to electronically and physically isolate working electrode (WE) and counter electrode (CE) or RE, which could improve electrolysis efficiency and avoid electrolysis of molten chloride salts at even high applied potentials.[21] It also provided a basal body integrating the CE with the RE. Moreover, compared with previous works,[2–15] the experimental apparatus contained neither sealable stainless steel reactor nor carbonaceous electrode or crucible, which could fundamentally eliminate their contamination to the electrolytic Si and the formation of $CO_x$ gases, also avoid carbon entering molten salt to reduce current efficiency due to electronic conduction.

The goal of the present work is to demonstrate the use of the experimental apparatus with the novel integrated cell as a simple and convenient tool for green electrochemical preparation of contamination-free Si from solid $SiO_2$ in the laboratory, while we hope that the findings reported in the work will benefit similar studies of other high-temperature electrolytes and the future design of the industrial cell capable of continuous operation, and the selection, monitor, and control of the process variables for optimal production.

## II. EXPERIMENTAL

### A. *Materials Preparation*

$SiO_2$ and $CaCl_2$ used in this work were analytical pure-grade reagents provided by Chinese Pharmaceutical Group Chemical Reagent Co., Ltd. The $SiO_2$ powder was loaded into a steel mold, and cylinder samples were obtained by using 1.56 MPa pressure. Each cylinder sample had about 5 mm diameter, 10 mm height and 0.12 g mass. They were sintered for 4 h at 1673 K under air, which gave them a certain porosity and strength. The $CaCl_2$ powder, used as the electrolyte, was dried under a vacuum of $< 133$ Pa for 8 h at 473 K for adequate dehydration.

### B. *Integrated Cell and Electrodes*

The integrated cell consisted of an YSZ tube with a closed end and three electrodes including CE, RE and Mo-$SiO_2$ WE. The YSZ tube (inner diameter 18 mm, outer diameter 20 mm and length 100 mm) contained 8 mol% $Y_2O_3$. After the YSZ tube was ultrasonically cleaned with alcohol, the porous annular Pt CE (area about 6.28 $cm^2$) and RE (area about 1.88 $cm^2$) were prepared according to the method provided in the literature.[26,28] The Pt leads (diameter 0.5 mm and length 800 mm) for the RE and the CE were each protected in a thin double-bore alumina tube.

The Mo-$SiO_2$ WE was prepared as follows. A 0.9-mm drill bit was used to punch the center of the sintered $SiO_2$ cylinder. One end of Mo wire (purity 99.99%, diameter 1 mm, length 800 mm), polished by fine sandpaper (2000 mesh), was tightly coated with a thin single-bore alumina tube and slid to leave a short length of bare wire through the central hole of the $SiO_2$ cylinder. The end of the bare wire was bent to prevent the $SiO_2$ cylinder from sliding into the molten salt during the process of lowering or electrolysis; the thin single-bore alumina tube covering the Mo wire tightened the upper end of the $SiO_2$ cylinder, which prevented the $SiO_2$ cylinder from moving up during the process of lowering.

### C. *Experimental Method*

Figure 1 shows schematically the experimental apparatus used in this work. Since the YSZ tube was shortened to only 100 mm in order to conveniently operate and isolate the atmosphere inside and outside the YSZ tube, the open end of the YSZ tube used as an electrolytic cell was connected with an extended alumina tube (inner diameter 21 mm, outer diameter 25 mm and length 600 mm) by high-temperature cement. The preliminary experimental result showed that the joint between the YSZ tube and the alumina tube was well sealed after sintering at 1173 K, and no leakage was found when it was immersed in water at room temperature. The thin double-bore alumina tube, covering the RE and the CE leads (Pt wires), was bound to the alumina tube with Cu wire (0.5 mm diameter). The electrode leads were respectively fixed with thin Pt wire

(0.3 mm diameter) to the corresponding RE and CE on the outer surface of the YSZ tube. With the alumina tube, the integrated cell loading the dried $CaCl_2$ powder (about 12 g) was put into the constant temperature zone of a high-temperature furnace. The heating element of the furnace was the SiC tube, the furnace tube (inner diameter 45 mm, length 100 cm) was alumina material, and an S type thermocouple was used to measure temperature of the furnace. The furnace was heated from room temperature to 723 K at a rate of 3 K/min and then continued to 1173 K at a rate of 4 K/min and held for 30 min. High-purity argon (purity 99.999%, $O_2$ ≤ 1.5 ppm), dried via a pair of the molecular sieves and the silica gel columns, was introduced into the YSZ tube with a flow rate of 10 mL/min throughout the experiment, while it was not until the experimental temperature was reached that the dried synthetic air (79% $N_2$, 21% $O_2$) with a flow rate of 300 mL/min was introduced into the furnace, that is outside the YSZ tube, to construct a $O^{2-}|YSZ|Pt|O_2$(air) RE. All potentials ($E$) in this work referred to the potential relative to the $O^{2-}|YSZ|Pt|O_2$(air) RE. Compared with the electrolytic cell in conventional crucible form, the amount of $CaCl_2$ powder used in the YSZ tube was very small, which was favorable for fully removing the moisture in the $CaCl_2$ during the process of heating, holding and pre-electrolysis.

Electrochemical tests were conducted with a three-electrode method using an electrochemical workstation (model: IviumStat.h, made in The Netherlands) connected to a computer. After holding the temperature, a Ni wire (purity 99.99%, diameter 0.5 mm) was used as cathode to pre-electrolyze for 90 min at − 1.8 V. After the Ni wire was removed, the Mo-$SiO_2$ WE was slowly lowered from the furnace top to the above of the melt level, and then the melt level was determined by the open circuit potential method to ensure that the whole $SiO_2$ cylinder was immersed in the melt.[28] Electrochemical behavior of the Mo-$SiO_2$ WE was investigated by cyclic voltammetry (CV) method. Based on this test, the electroreduction of solid $SiO_2$ was studied at a lower overpotential by the chronoamperometry (potentiostatic electrolysis) method.

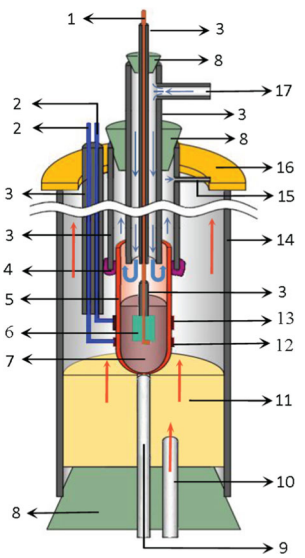



Fig. 1—Experimental apparatus used in this work.

The synthetic air in the furnace was cut off after the measurements were finished. The Mo-$SiO_2$ WE in the YSZ tube was lifted above the melt level and cooled to room temperature with the furnace under the protection of argon. The appearance of the YSZ tube was first observed with the naked eye. And each electrolyzed cylinder was soaked for 30 min in a beaker with a small amount of distilled water. One could observe whether gas bubbles were formed in the water, and then we used pH paper to detect the acidity and alkalinity of the water. The electrolytic products were further washed ultrasonically with distilled water for 3 min and then dried for 20 h in the vacuum at 343 K. The dried cylinder sections were characterized by scanning electron microscopy (SEM, Nova NovaSEM400) with energy-dispersive x-ray spectroscopy (EDS, INCA IE 350PentaFET X-3EDS). If necessary, the section of the Mo current collector was also observed under SEM-EDS. Since the mass of the $SiO_2$ cylinder was small to only 0.12 g, phase detection was not carried out by x-ray diffraction (XRD) after electrolysis. For comparison with the electrolytic products, morphology observation, composition analyses and phase detection for the $SiO_2$ raw powder and the sintered $SiO_2$ cylinder were conducted with SEM-EDS and XRD, respectively.

## III. RESULTS AND DISCUSSION

### A. *Cv*

Figure 2 shows a typical cyclic voltammogram of the Mo-$SiO_2$ WE in the $CaCl_2$ melt. A scan is performed with a rate of 50 mV/s in a potential range of − 0.5 to − 1.8 V. It is clear that a reduction peak $C_1$ first appears. Notably, the potential (− 0.73 V) of peak $C_1$ is close to the theoretical potential (− 0.75 V) of $MoO_3$ reduction to Mo,[30] indicating that the peak $C_1$ corresponds to the reduction of $Mo^{6+}$ to Mo in the melt (Eq. [2]):[31]

$$Mo^{6+} + 6e \rightarrow Mo \qquad [2]$$

There are two reasons for the formation of $Mo^{6+}$ in the melt. One is that, when the Mo-$SiO_2$ WE is lowered, the Mo wire reacts with the residual oxygen in argon at high temperatures to form a small amount of $MoO_3$ (melting point 1068 K, boiling point 1428 K), some of which volatilize, and some adhere to the surface of the Mo wire in the melt. A graphite electrode or crucible was often used in previous works,[2–15] which could protect Mo from oxidation by the atmosphere; hence, no reduction peak of $Mo^{6+}$ was observed. The other is that Fig. 2 shows a large oxidation current when the potential scan is initially performed from − 0.5 V, indicating that the Mo wire in the melt may have been electrochemically oxidized to $Mo^{6+}$. Obviously, the peak $A_1$ with the potential of about − 0.65 V corresponds to the oxidation of Mo to $Mo^{6+}$.

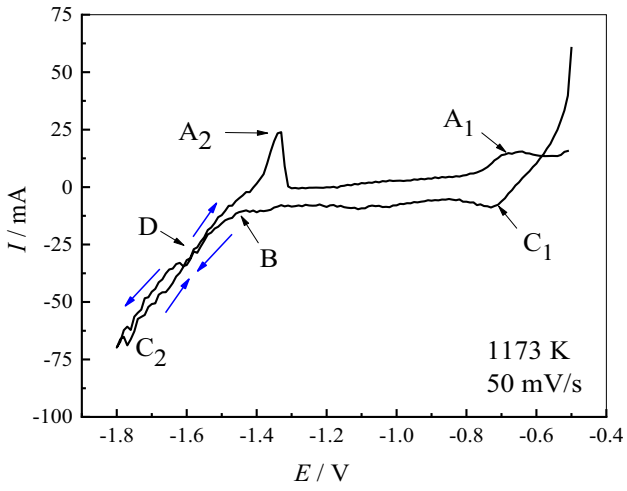



Fig. 2—Typical cyclic voltammogram of the Mo-$SiO_2$ WE in $CaCl_2$ melt (RE: $O^{2-}$|YSZ|Pt|$O_2$(air)).

As the scan continues forward, the current changes little prior to position B where the onset potential of a new peak $C_2$ is about − 1.45 V. The peak $C_2$ is incomplete and most likely to correspond to the $SiO_2$ reduction. However, the reduction potential of $SiO_2$ at 1173 K is − 1.77 V from thermodynamic theory calculation.[30] It is inferred that the onset potential of the peak $C_2$ corresponds to $SiO_2$ reduction on the Mo to form an alloy with the Si activity of less than unity, which results in an advance of the $SiO_2$ reduction potential to − 1.45 V. Figure 3 shows a SEM image of a cross section of the Mo wire after potentiostatic electrolysis. The EDS analyses of relevant position points in Fig. 3 are listed in Table I. As expected, a Si-Mo alloy layer formed at the edge of the Mo wire is found. As the position moves from the edge to the center of the Mo wire, the Si content in the alloy decreases and Mo content increases, indicating that Si gradually alloyed with Mo from the surface to the center of the Mo wire. Inexplicably, the formation of alloy between Mo and Si was rarely noted in previous works.

A crossover of the forward and reverse curve at position D (around − 1.60 V) is observed, indicating the increases of the total length of 3PIs and also the area of the Si/$CaCl_2$ interface where the new phase Si nuclei are generated.[2,29] Furthermore, the reoxidation part exhibits a sharp decrease in current after peak $A_2$, which is typical of a stripping effect due to the depletion of the product reduced during the forward (negative) scan. Obviously, the $A_2$ with peak potential of − 1.34 V can be attributed to the reoxidation of Si.

### B. *Potentiostatic Electrolysis*

Under the experimental conditions, it was found that the $SiO_2$ could be reduced at − 1.45 V according to Fig. 2, and the crossover of the forward and the reverse curve occurred at − 1.6 V. Therefore, the cross-potential of − 1.6 V was used for electrolysis for 3 ks, 5ks and 6 ks, respectively. The preparation of Si and the change of its morphology from electrolysis of solid $SiO_2$ were

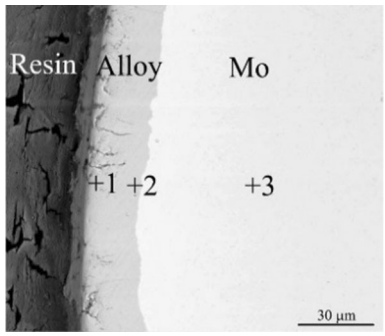



Fig. 3—SEM image of Mo wire section after potentiostatic electrolysis for 5 ks at 1173 K and − 1.8 V (RE: $O^{2-}$|YSZ|Pt|$O_2$(air)).

studied by using the integrated cell in this experiment. Figure 4 shows the current curves with time for $SiO_2$ electrolysis at − 1.6 V and 1173 K; the current curves are similar.

The YSZ tube cooled to room temperature was found to exhibit obvious moisture state on the outer surface position below the melt level in the YSZ tube after the electrolysis for different times, but no crack or melt leakage was observed with the naked eye. Moisture is a clear sign of water absorption of $CaCl_2$ penetrating into the YSZ membrane. It is tentatively speculated that the permeability of $CaCl_2$ enhanced in the inner surface of the YSZ tube in this work. This will be further explained later.

For each cylindrical sample soaked in the water after electrolysis for different times, no obvious bubble was observed, and the water was neutral according to pH paper, indicating that there was no Ca metal or Ca-Si alloy in the electrolytic products, which is consistent with the CV result. After further ultrasonic washing and drying, a brown product layer can be observed in the sections of the three cylinders with different electrolysis times, which is obviously different from the white $SiO_2$ raw material. The presence of brown color indicates that the $SiO_2$ has been electrolyzed to Si in this experiment.[29,32] It is believed that solid $SiO_2$ is reduced to Si at the cathode (Eq. 1); the oxide ions produced from the solid structure migrate in the melt and solid electrolyte and are oxidized to $O_2$ gas at the anode outside the YSZ tube (Eq. [3]).

$$2O^{2-} \rightarrow O_2 + 4e \qquad [3]$$

However, an obvious white layer could also be found in the outer region of the cylinder section far away from the Mo current collector for electrolysis time of 3 ks and 5 ks, respectively. An integral calculus for the current curve with electrolysis of 6 ks was performed. According to the Faraday law, the reduction rate with electrolysis of 6 ks was only 54.4% in theory even if the current efficiency was assumed as 100%, indicating that the electroreduction of solid $SiO_2$ cylinders was also incomplete within the given time of 6ks because of the small electrolysis current. It could be expected that the whole $SiO_2$ cylinder was reduced to Si by an extension of the electrolysis time.

**Table I. EDS Analyses of Relevant Position Points in Fig. 3**

| Points | Atomic Fraction (Pct) | |
|---|---|---|
| | Mo | Si |
| 1 | 14.35 | 85.65 |
| 2 | 22.57 | 77.43 |
| 3 | 100 | — |

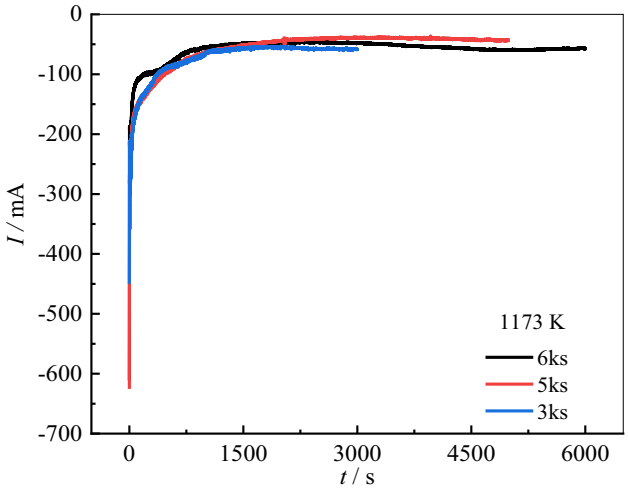



Fig. 4—Current curves with time for solid $SiO_2$ electrolysis in $CaCl_2$ melt at − 1.6 V (RE: $O^{2-}$|YSZ|Pt|$O_2$(air)).

Here, the morphology and composition of the brown products obtained were mainly analyzed with SEM-EDS detection, whereas the alloy formed on the Mo current collector and white unreduced layer was not considered.

The morphology of the $SiO_2$ raw powder and the sintered cylinder was first observed under SEM, as shown in Fig. 5. The $SiO_2$ raw powder is loose (Fig. 5(a)); the difference in the particle size is large and generally between 5 and 50 $\mu$m. The particles in the cross section of sintered cylinder become tight. The number of small particles increases, and they easily adhere on the surface of large particles (Fig. 5(b) and 5(c)), indicating that some large particles of $SiO_2$ raw powder are crushed and become small after pressing and sintering. The particles experienced a slight softening-melting phenomenon after sintering and even adhered to each other slightly (Fig. 5(c)), but they could still maintain their shapes with different sizes. Moreover, the $SiO_2$ raw powder and the sintered cylinder were found to be pure $SiO_2$ crystal phase by XRD and EDS detection (not shown here), with no change in the composition.

Figure 6 shows SEM images of the brown products from electrolysis for 3 ks. The morphology is completely different compared with the sintered $SiO_2$ particles (Fig. 5(b) or 5(c)). The brown products obtained are loose spongy particles (Fig. 6(a)); most of them are large spherical particles of about 1 $\mu$m diameter at higher magnification (Fig. 6(b)). Some large spherical particles aggregate with each other, and some small particles within about 300 nm diameter are not uniformly gathered around the aggregates. Moreover, some beads with diameters of 100 to 300 nm or rod-like substances of $< 1.5$ $\mu$m in length grow on the surface of some large spherical particles (Fig. 6(b)). It can be inferred that a large change of the morphology of the particles before and after electrolysis is attributed to the electroreduction of $SiO_2$ particles and the physicochemical interaction between the $SiO_2$ particles and the melt. The morphology of electrolytic products has little relation with that of the sintered $SiO_2$ particles. The EDS analyses of the relevant position points in Fig. 6(b) are listed in Table II. The EDS detection (Table II) shows that the Si content of each position in the electrolytic products in general is higher than that in the sintered $SiO_2$ particles before electrolysis, while the O content in the spherical particles is higher than that in the small particles after electrolysis. Occasionally, a small amount of $CaCl_2$ or CaO is found. The $CaCl_2$ is the residue after washing, while the CaO is generated because of hydrolysis of $CaCl_2$ or during the electroreduction of $SiO_2$, and it can further react with solid $SiO_2$ to form an intermediate of $CaO{\cdot}SiO_2$.[5,28,33,34]

The color change and EDS analyses of the cylinder before and after electrolysis show that the large and small particles evolved from the original $SiO_2$ powder have been partially reduced. As mentioned earlier, the partially reduced spherical particles and small particles, together with beads and rod-like substances, compose the aggregates; their distributions are also uneven in the aggregates. It can be considered that the beads and the rod-like substances with higher Si content grew from Si nuclei. The Si nuclei could be generated at − 1.6 V, which seems to be consistent with the CV result. Combined with the 3 PIs theory on the electroreduction of solid $SiO_2$ in molten salt,[17,29] it is inferred that a conductive Si network has been initially formed in the aggregates (especially including small particles and rod-like substances), which can act as a new network current collector to promote further reduction of solid $SiO_2$.

Figure 7 shows the SEM images of brown products from electrolysis for 5 ks. Compared with Fig. 6, the morphology of the products also changes greatly. Although the products are also composed of small particles and rod-like substances or wires, the diameter and length of rod-like substances increase, and the number increases significantly. In contrast, the number of spherical particles with higher O content decreases significantly. Rod-like substances grow longitudinally on the surface of spherical particles, some of which continue to grow into wires; some tend to merge into flakes. The rods or the wires are of varying diameter. The gaps in the products also increase obviously, indicating that the removal amount of O in the $SiO_2$ cylinder increases. The EDS analyses of the relevant position points in Fig. 7(b) are listed in Table III. The EDS detection (see Tables II and III) shows that the main constituent in small particles or rod/wire-like substances is Si, and Si contents in some positions are

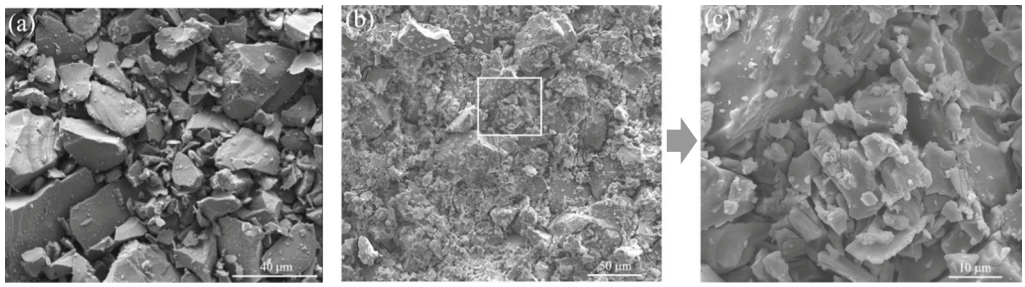



Fig. 5—SEM images of $SiO_2$ raw powder (*a*) and sintered cylinder section (*b*) and local magnification (*c*).

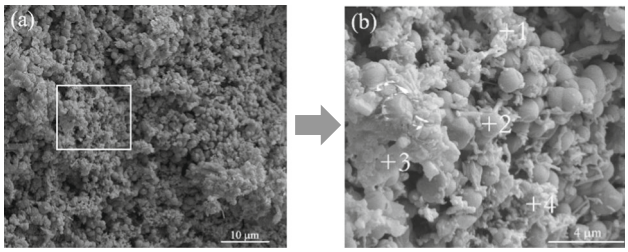



Fig. 6—SEM images of cylinder section after electrolysis for 3 ks (*a*) and local magnification (*b*).

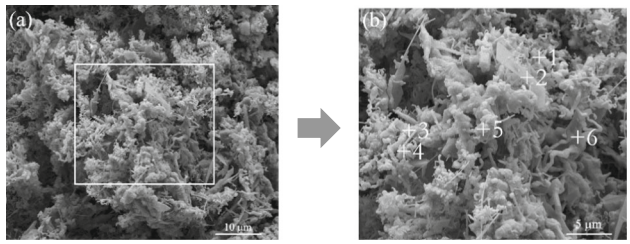



Fig. 7—SEM images of cylinder section after electrolysis for 5 ks (*a*) and local magnification (*b*).

**Table II. EDS Analyses of Related Position Points in Fig. 6(b)**

| | Atomic Fraction (Pct) | | |
|---|---|---|---|
| Points | Si | Ca | O |
| 1 | 69.80 | — | 30.20 |
| 2 | 49.17 | — | 50.83 |
| 3 | 86.42 | — | 13.58 |
| 4 | 66.80 | 5.87 | 27.32 |

even as high as 100 at%. By contrast, some near-spherical particles contain more unreduced $SiO_2$. It is known that many experiments[29,32,35,36] on the electroreduction of solid $SiO_2$ in molten salt were performed in a sealed stainless steel reactor with argon flow, and even with the use of a carbonaceous electrode or crucible, the surface of the reduced Si was still found to be covered with a thin layer (thickness around 10 nm) of amorphous $SiO_2$, which was generally attributed to slight oxidation of Si in Ar atmosphere at high temperatures or passivation during washing at room temperature, indicating that it is very difficult to obtain high purity Si with very low oxygen content only by electrolysis in Ar atmosphere.

In view of the 3 PIs theory, combining the morphology of the products from electrolysis for 3 ks, it can be speculated that the solid $SiO_2$ raw powder was spheroidized into new particles under the electrolysis and the physicochemical interaction between the $SiO_2$ particles and the melt, and then fine Si nuclei grew on the surface of spherical particles, formed Si network current collectors along with small particles with low oxygen content and continued to grow longitudinally into rod-like substances, and finally into Si wires or flakes. Meanwhile the network current collectors further extended until the original $SiO_2$ disappeared because of complete reduction.

Figure 8 shows SEM images of brown products from electrolysis for 6 ks. The morphology of the products (Fig. 8(a) and 8(b)) is basically the same as that in Fig. 7. It is mainly composed of small particles, wires and flakes. The wire is considered to grow from the rod-like substances mentioned earlier; the difference in diameter between the two is small, but the longitudinal length of the former increases. It is believed that most of rod-like substances have grown into the wires, and there are some sites where the rod-like substances merge into each other to form larger flakes under specific conditions (Fig. 8(b)). The number of spherical particles is small, indicating a further increase in reduction. The EDS sum spectrum (Fig. 8(c)) shows that Si content in the brown products exhibited in Fig. 8(b) is $> 95$ at% locally, and the content of O or Ca is very low and can be further reduced with adequate washing. The Ca content is slightly higher than the O content, which may be due to an error in the EDS detection or Cl not being included. Combined with the above-mentioned detection on the electrolytic products soaked in water, it can be concluded that no Ca metal or Ca-Si alloy was formed during the electrolysis. Furthermore, no Zr or Y from the YSZ tube was found in the products either. Therefore, under completely carbon-free conditions, the Si specimens with local high purity can be obtained in the YSZ tube (where the gas is under argon atmosphere) even if the gas outside the YSZ tube is under the air in this work, indicating that the YSZ tube can not only act as an oxygen ion isolation membrane, but also has a good atmosphere isolation function at high temperatures, which also proves that the joint part between the YSZ tube and the extended alumina tube

has a good seal. It is possible that the sealable stainless steel reactor and carbonaceous anode or crucible used in previous works[2–15] are not essential during the electroreduction of solid $SiO_2$ to Si. The experimental apparatus used in this work not only meets the requirement of preventing the oxidation of the active Si, but also eliminates fundamentally the potential contamination of the Si product caused by the stainless steel reactor, carbon and even Ca metal or Ca-Si alloy, which provides the possibility to establish a green technique for preparation of high-purity Si and the release of $O_2$ gas by electrolysis of solid $SiO_2$ in $CaCl_2$ melt. Moreover, the longer the electrolysis time is, the looser the reduction products, the more and longer the Si wires, and the higher the Si content. Of course, still there are several other potential contamination sources such as the alumina tube and the Mo wire electrode in this work. To prepare Si from solid $SiO_2$ without being contamination-free in the $CaCl_2$ melt and obtain high purity Si in the succeeding work, referring to the related findings,[4,16] it can be considered to replace the $Al_2O_3$ tube touching the $SiO_2$ cylinder with the Si or $SiO_2$ tube and to replace the Mo wire electrode with the Si rod. It is also necessary for the electrolytic products to be cleaned fully.

Notably, the morphology of the Si product in this work is obviously different from that in the works of Zhao[14] and Lai et al.[15] in $CaCl_2$ melt. They used the SOM process with a two-electrode method and applied voltages (3.2 to 4 V) to electrolyze solid $SiO_2$ at $> 1273$ K. Si particles, rather than the Si wires and flakes in this work, were obtained, which might be attributed to the larger voltages and higher temperatures in their works.

**Table III. EDS Analyses of Related Position Points in Fig. 7(b)**

| | Atomic fraction / % | | | |
|---|---|---|---|---|
| Points | Si | Ca | O | Cl |
| 1 | 64.23 | 3.65 | 29.26 | 2.86 |
| 2 | 81.97 | — | 18.0 | — |
| 3 | 100.00 | — | — | — |
| 4 | 64.84 | 6.00 | 29.17 | — |
| 5 | 79.74 | — | 20.26 | — |
| 6 | 100.00 | — | — | — |

The morphology of the Si product may be controlled by applying different potentials. Furthermore, Ca metal or Ca-Si alloy could also be obtained at large voltages, which would adversely affect the electrochemical reduction of solid $SiO_2$.[29]

The stability of the YSZ tube is also an important issue. Due to the different cell configuration, the moisture phenomenon that appeared in this work is not seen in the works of Zhao[14] and Lai et al.[15] Notably, the electrochemical behavior of $Fe_2O_3$ dissolved in $CaCl_2$-NaCl melt was studied by using the same cell as in this work (except for the size of the YSZ tube).[28] Unlike this work, however, no moisture phenomenon was observed on the outer surface of the YSZ tube, and the $O^{2-}|ZrO_2|Pt|O_2$ (air) RE worked normally. A considerable number of studies on using the SOM process with a two-electrode method to prepare metals such as Mg, Si and Al from respective dissolved oxides show that the YSZ tube in molten halides presents a certain instability.[37–41] The source of instability lies in the leaching of stabilizer $Y_2O_3$ in solid solution, which induces the transformation of $ZrO_2$ from cubic to monoclinic phase, resulting in YSZ tube cracking or even breaking. Moreover, the electronic conductivity of the system exacerbates this instability. Considering the stability of the YSZ tube, the content of dissolved $SiO_2$ or $Al_2O_3$ in the melt should not be too high.[37,38] It is also worth noting that the role of the dissolved CaO in YSZ stability seems contradictory in the literature[38, 40, 41] , which may be attributed to the difference of the melt. However, the addition of $Y_2O_3$ or $YF_3$ to the melt was found to alleviate or completely prevent the YSZ tube degradation resulting from $Y_2O_3$ chemical leaching.[37–41] Combining these findings,[37–41] the stability of the YSZ tube in the melt can be preliminarily estimated in this work. It is well known that $CaCl_2$ exposed to air is easily hygroscopic at room temperature. The literature has shown that the $CaCl_2$ could penetrate into and remain in the grain boundaries and pores left in the $ZrO_2$ grains after $Y_2O_3$ leaching.[37,40] After the experiment, it is the hygroscopicity of the residual solidified $CaCl_2$ in the YSZ membrane that results in the moisture observed on the outer surface of the YSZ tube. Since the applied potential has little negativity, the time is short at high temperature and the solubility of $SiO_2$ or $CaO{\cdot}SiO_2$ in the melt is small,[34] it can be understood that no crack or salt leakage in the

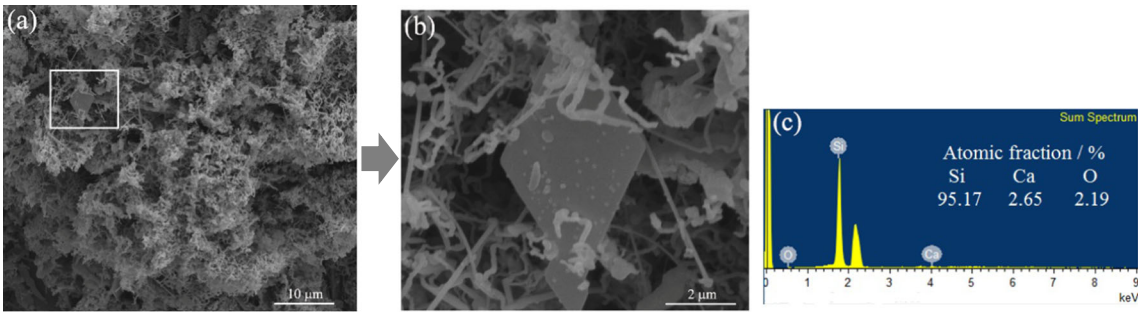



Fig. 8—SEM images (*a*) of cylinder section after electrolysis for 6 ks and local magnification (*b*) and its EDS sum spectrum (*c*). (The peak of unlabeled element in the sum spectrum is attributed to that of gold).

YSZ tube was found. Referring to the literature findings,[37–41] some measures to modify the melt, such as adding $Y_2O_3$ ($YF_3$ or $YCl_3$) or adjusting the optical basicity of oxides ($SiO_2$ and CaO), can be taken, and if direct touching of the solid $SiO_2$ with the YSZ tube is avoided during the experiment, the degradation of the YSZ tube in the $CaCl_2$-CaO melt can be expected to be alleviated or prevented completely in future preparation of high-purity Si using the integrated cell.

## IV. CONCLUSIONS

A novel integrated cell with $O^{2-}|YSZ|Pt|O_2(air)$ RE and CE was constructed using a short YSZ tube connected to a long-extended alumina tube by high-temperature cement. In the experimental apparatus containing the integrated cell without a stainless steel reactor or any carbonaceous materials, Si specimens with local high purity > 95 at% were successfully obtained by electrolysis of solid $SiO_2$ in the $CaCl_2$ melt at − 1.6 V and 1173 K, and $O_2$ gas was simultaneously released. The CV test indicates that the reduction peak potential of $Mo^{6+} \rightarrow Mo$ is − 0.73 V and the onset of the reduction peak potential of $SiO_2 \rightarrow Si$ is − 1.45 V in this work. The potentiostatic electrolysis at low overpotential shows that the morphology of the products underwent an evolution from $SiO_2$ raw powder with different sizes to particle aggregates with partial reduction, and then Si nuclei formed on the surface of spherical particles, further growing longitudinally into rod-like substances and finally into Si wires or flakes. The morphology of electrolytic products has little relation with that of $SiO_2$ raw powder and may be controlled by applying different potentials. Furthermore, the longer the electrolysis time is, the higher the purity of Si on the whole. The present work demonstrates the feasibility of green electrochemical preparation of Si from solid $SiO_2$ by using the experimental apparatus containing the novel integrated cell with $O^{2-}|YSZ|Pt|O_2(air)$ RE under complete carbon-free conditions.

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